

CRYPTOGRAPHY & COMPUTER SECURITY – EXTENSIVE MASTER RESEARCH PACK

PART I: CRYPTOGRAPHY

1. Foundations of Cryptography Cryptography is the scientific study of securing communication and data from adversaries. Its foundational goals—Confidentiality, Integrity, Authentication, and Non-repudiation—form the basis of secure digital systems. Modern cryptography relies heavily on mathematical hardness assumptions such as factoring (RSA) and discrete logarithms (DH, ECC). Adversarial models include passive eavesdroppers, active man-in-the-middle attackers, and adaptive chosen-ciphertext attackers.

2. Classical Cryptography Early ciphers include substitution and transposition systems such as Caesar, Vigenère, Playfair, and Hill cipher. Despite being historically important, these are vulnerable to frequency analysis, Kasiski examination, and brute-force search. Their main contribution is foundational: inspiring modern cryptosystems based on more complex transformations.

3. Modern Symmetric Cryptography Symmetric systems use a single shared key. Block ciphers like DES, AES, Blowfish, Twofish operate on fixed-size blocks using substitution-permutation networks. Stream ciphers (RC4, ChaCha20) generate keystreams XORed with plaintext. Secure modes of operation include CBC, CFB, OFB, CTR, and authenticated encryption modes like GCM. Key management and secure random number generation (CSPRNGs) are critical for safety.

4. Asymmetric Cryptography Public-key cryptography solves key distribution problems using mathematical one-way functions. RSA relies on integer factorization, Diffie-Hellman on discrete logs, and ECC on elliptic-curve discrete logs. Applications include key exchange, signatures, and secure email. Modern systems prefer ECC due to smaller keys and high security.

5. Hash Functions and MACs Cryptographic hash functions (SHA-2, SHA-3) provide fixed-length digests and are essential for integrity. MAC algorithms (HMAC, CMAC) combine hashing or block ciphers with secret keys. Collision resistance, preimage resistance, and avalanche effect define their strength.

6. Digital Signatures and PKI Digital signatures provide authenticity and non-repudiation using RSA, DSA, or ECDSA. PKI manages trust through Certificate Authorities, CRLs, and OCSP. Secure certificate validation prevents impersonation attacks.

7. Advanced Cryptography Zero-knowledge proofs allow one party to prove knowledge without revealing it. Homomorphic encryption enables computation on encrypted data. Post-quantum cryptography explores quantum-resistant schemes such as lattice-based algorithms. Threshold cryptography, MPC, and blockchain cryptographic primitives (Merkle trees, mining puzzles) support decentralized trust.

8. Network & Protocol Cryptography Protocols like SSL/TLS, IPsec, SSH, and Kerberos use cryptography to secure communication. TLS handshake negotiates keys using asymmetric algorithms and encrypts sessions using symmetric ciphers. Wi-Fi security evolved from WEP to WPA2 and WPA3 incorporating AES-CCMP and SAE.

9. Cryptanalysis Attacks include brute-force, differential and linear cryptanalysis, side-channel attacks (timing, power analysis), and meet-in-the-middle. Modern systems must mitigate such vulnerabilities using constant-time implementations and hardened key schedules.

PART II: COMPUTER SECURITY

1. **Introduction to Information Security** Security protects systems from threats, vulnerabilities, and risks. Frameworks such as the CIA triad and threat modeling (STRIDE, DREAD) guide system design. Policies define acceptable behavior and access control expectations.
2. **Operating System Security** OS security manages authentication, process isolation, memory protection, and permissions. Access control models include DAC, MAC, and RBAC. Linux systems rely on file permissions, SELinux, and AppArmor. Windows uses NTFS ACLs, Active Directory, and Group Policies.
3. **Network Security** Network defense uses firewalls, IDS/IPS, NAT, VPNs, and segmentation. Attacks include DDoS, MITM, spoofing, ARP poisoning, and port scanning. Secure network design isolates critical assets and uses encrypted tunnels.
4. **Application Security** Applications must defend against the OWASP Top 10 threats such as SQL Injection, XSS, CSRF, broken authentication, and insecure deserialization. Secure development lifecycle (SDL), code reviews, and fuzz testing improve resilience.
5. **Web Security** HTTPS safeguards web communication. Session management, secure cookies, and CSRF tokens protect users. Common threats include injection attacks, privilege escalation, and server misconfigurations. TLS certificates ensure authenticity.
6. **Malware & Cyber Threats** Malware includes viruses, worms, trojans, ransomware, rootkits, and botnets. Cyber threat actors range from script kiddies to APT groups. Defensive approaches include antivirus, sandboxing, behavior analysis, and endpoint detection and response (EDR).
7. **Incident Response & Digital Forensics** Incident response involves identification, containment, eradication, recovery, and lessons learned. Forensics includes log analysis, disk imaging, timeline reconstruction, and chain-of-custody procedures. SIEM systems aggregate security logs.
8. **Cloud Security** Cloud environments follow a shared responsibility model. Security covers IAM, encryption at rest/in transit, network security groups, container security, virtualization threats, and misconfiguration risks. Tools include CSPM, CWPP, and Kubernetes RBAC.
9. **Identity & Access Management** IAM governs users and permissions. Authentication methods include passwords, biometrics, MFA. Authorization uses roles, policies, and tokens. Protocols include OAuth 2.0, SAML, and OpenID Connect.
10. **Physical Security** Security controls include biometric access, surveillance, hardware locks, environmental controls, and secure device disposal. Tamper detection protects servers and networking equipment.
11. **Governance, Risk & Compliance** Frameworks such as ISO 27001, NIST SP 800-53, COBIT, GDPR, HIPAA, and PCI-DSS govern cybersecurity programs. Risk assessment identifies impact and likelihood to prioritize controls.
12. **Emerging Trends** IoT security addresses device authentication and firmware integrity. Cyber-physical systems protect industrial and transportation infrastructure. AI enables both defensive automation and offensive attacks. Quantum computing threatens current cryptographic standards while inspiring quantum-resistant systems.

This document is designed as a research pack and foundation for exam preparation or policy development.